

Multi-agent interaction and enhanced safety in autonomous overtaking with backward reachability analysis

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ABSTRACT

Safe and efficient autonomous driving in complex multi-agent traffic scenarios requires proactive decision-making to anticipate human driver behavior while remaining robust against uncertainty in their actions. A key challenge lies in enabling collision-free navigation without relying on overly conservative motion planning strategies. Hamilton-Jacobi Backward Reachability Analysis (BRA) offers a powerful framework for safety verification in autonomous systems by defining a metric that characterizes the set of states from which collisions become inevitable, even under optimal control. In this work, the computational burden of computing the Backward Reachable Tube (BRT) via the Level Set Method is mitigated through a targeted grid selection strategy tailored for realistic driving scenarios, where safety is integrated into the Model Predictive Control (MPC) framework through a dual-layered approach. First, a minimally interventional reachability-based constraint is introduced and activated only in near-unsafe configurations that could lead to unavoidable collisions to preserve planning flexibility. Second, a novel attentive-level module based on reachability analysis is proposed, enabling continuous risk assessment in the presence of multiple human-driven vehicles. To provide a complete evaluation of the algorithm's effectiveness, simulations are conducted both with the safety-preserving constraint enabled and disabled, revealing potential collisions and highlighting the contribution of the proposed algorithm. The combined effect of the attentive layer and the reactive safety-preserving constraint is demonstrated through autonomous multi-agent overtaking scenarios on two-lane roads involving both preceding vehicles and oncoming traffic, highlighting improved safety and performance with respect to baseline methods.

1. Introduction and state of the art

In an ideal setting, an autonomous vehicle would simply follow a pre-defined path computed at the start of the trip. However, real-world driving entails numerous complexities, including road boundaries, dynamic agents, cyclists in urban environments, merging traffic on highways, and sudden pedestrian appearances. The behavior of surrounding drivers, whether cooperative or adversarial, can significantly impact safety (Zhao et al., 2020), as well as the considered driving scenario.

The uncertainty introduced by human behavior, combined with sensor limitations and dynamic environments, adds a significant layer of complexity to autonomous decision-making and safety assurance (Cruz-Maya & Europe, 2020). To address this, safe control strategies aim to bound such uncertainty, ensuring the ego vehicle remains within safe and predictable operating conditions.

One foundational approach in motion planning is the use of artificial potential fields (APFs), introduced by Khatib (1986). APFs guide vehicles using attractive forces toward goals and repulsive forces away from obstacles. While intuitive and effective in simplified environments, these methods often suffer from local minima and typically neglect the underlying system dynamics (Song et al., 2023). To overcome these limitations, researchers have proposed enhancements such as trajectory sampling (Hegedes et al., 2020), velocity potential fields (Xie et al., 2022), and integration with model predictive control (MPC) (Dixit et al., 2020), which introduce greater adaptability and dynamic feasibility.

To better incorporate vehicle dynamics and enforce formal safety constraints, control barrier functions (CBFs) have emerged as a promising alternative (Ames et al., 2019). CBFs encode safety as state constraints, ensuring that the system remains within certified safe sets. These are often employed alongside Lyapunov-based techniques to

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balance safety and stability. However, real-world applications often face practical issues such as constraint infeasibility due to limited control authority or conflicting objectives (Garg et al., 2023). To address this, researchers have developed hierarchical formulations (Xie et al., 2024), quadratic programming methods, and combinations with reachability-based analysis for more robust constraint handling.

Complementing these model-based approaches, learning-based methods show flexibility in complex and uncertain environments (Paparazzo et al., 2025). Techniques such as behavioral cloning and reinforcement learning allow autonomous systems to generalize from data and learn adaptive policies (Teng et al., 2023), with human behavior models used to achieve near expert decisions on urban roads (Yang et al., 2021). While effective in typical driving conditions (Song et al., 2023), these methods tend to struggle with rare or adversarial corner cases, raising concerns about robustness. Enhancements like hierarchical learning (Gurses et al., 2023; Lu et al., 2022), curriculum learning (Liu et al., 2022), risk-aware training (Rezaee et al., 2021), and hybrid architectures (Yang et al., 2022) aim to improve performance under uncertainty and enhance reliability.

Beyond learning from experience, game-theoretic frameworks offer a way to model interactions among multiple agents. These frameworks treat each driver or vehicle as a strategic player whose decisions influence others, making them especially relevant for dense, multi-agent scenarios. General-sum games capture the competitive and cooperative nature of real-world interactions (Censi et al., 2021). Despite their theoretical appeal, game-theoretic models often face scalability challenges and rely on assumptions about agent rationality or intent (Zanardi et al., 2023). Solutions such as hierarchical decomposition (Fisac et al., 2018), linear-quadratic approximations (Fridovich-Keil et al., 2019), and homotopy-based tactical planners (Khayyat et al., 2024) aim to reduce the computational burden while retaining predictive power. This framework can be further enhanced by combining learning methods with game theoretic formulations, by computing equilibrium oriented policies with Nash Q learning (Zhou et al., 2020) or cooperative mixed traffic using contrastive multi agent deep reinforcement learning (Qian et al., 2024).

For formal verification and proactive risk assessment, reachability analysis provides safety guarantees by using a conservative approach. By computing the set of all states a system may enter over time, reachability methods can provide guarantees on safety and goal attainment (Bansal et al., 2017). Forward reachability is used to predict the evolution of other agents (Coskun, 2021), while backward reachability is typically used to identify the states that must be avoided to ensure safety (Leung et al., 2020). In the context of multi-agent systems, these techniques are especially valuable in differential game formulations (Bansal et al., 2017), which naturally integrate multiple interacting agents. Nonetheless, the high computational cost of reachability remains a barrier to real-time deployment (Mitchell, 2020). Toolkits such as *helperOC* (Herbert et al., 2020) and *hj_reachability* (StanfordASL, 2024), as well as decomposition strategies (Chen et al., 2016), have been developed to address scalability. More recently, hybrid methods combining reachability with learning-based approaches show promise in handling high-dimensional and black-box dynamics (Bansal & Tomlin, 2020; Chilakamuri et al., 2024).

Related work. In autonomous driving, some maneuvers involve higher danger and require conservative assumptions to be performed. Thus, overtaking is selected as the testing scenario due to its intrinsic risk, which makes the design of robust, but not conservative, control strategies particularly challenging. Recent research focuses mainly on high-level decision policies, human behavior modeling and safety assurance. Most motion planning frameworks remain optimization-based, efficiently combining these aspects within predictive control formulations. However, achieving real-time feasibility while ensuring safety of the overtake remains a significant challenge. To reduce computational effort, lateral and longitudinal dynamics are often decoupled under the assumption of small road curvature, which holds for highway scenarios.

In this context, Lattarulo et al. (2018) presents a linear MPC approach that dynamically adapts to surrounding vehicles by adjusting both speed and trajectory to prevent potential collisions. Building on this, Tariq et al. (2022) enhances computational tractability by reformulating the overtaking optimal control problem (OCP) as a mixed-integer program that accounts for limited sensing capability and achieves real-time execution. Focusing on safety guarantees, Wang et al. (2024) introduces elliptical constraints to represent surrounding agents and employs road-edge barrier functions to respect boundaries. Similarly, Ammour et al. (2022) combines MPC with a sigmoid barrier to effectively avoid collisions across multiple overtaking scenarios; however, the computational complexity scales poorly with the number of surrounding vehicles in multi-agent settings. To achieve safety guarantees, Yuan et al. (2024) proposes a varying-level CBF integrated within an MPC with a simplified dynamics to ensure a safe-by-construction trajectory that remains feasible both for overtaking and returning behind the lead vehicle. Other works have explored probabilistic and game-theoretic formulations to improve interaction modeling. Németh et al. (2019) develops a graph-based optimal route and speed selection algorithm that optimizes velocity and minimum safe distance through a probabilistic framework. To explicitly capture interactive decision-making, Yu et al. (2024) employs a Stackelberg game formulation combined with robust MPC, ensuring collision avoidance through geometric constraints based on the autonomous vehicle's speed and position, although longitudinal control is neglected. Complementarily, Li et al. (2018) models driver behavior through seven basic actions determined by a reward function that balances safety, constraint violations, and speed maximization, providing a benchmark for evaluating control algorithms in multi-agent settings. Finally, Wang et al. (2020) introduces a reachability-based approach that formally incorporates vehicle dynamics into safety-preserving constraints. Pairwise interactions are efficiently managed in highway scenarios, however, extending the framework to multiple agents remains an open challenge.

Contributions. Recent state of the art reviews (Dixit et al., 2018; Lodhi et al., 2023; Ortega et al., 2024), converge on finding some main weaknesses in current overtaking algorithms. Mainly, on board computation is heavy, especially for nonlinear MPC and interaction modeling in multi agent traffic and human factor realism remains limited. This work addresses these gaps by integrating BRA into a real-time MPC and by adding a safety layer capable of handling several human driven vehicles, testing the framework in realistic multi-agent traffic by ensuring safety during autonomous overtaking on bidirectional scenarios, thereby bringing research closer to real world applicability.

In particular, to handle uncertainty in human behavior, BRA is used to generate safety-preserving constraints under bounded rationality assumptions. A scalable and tractable implementation is achieved through a decomposition of the high-dimensional BRTs, supported by symmetry exploitation and scenario-specific grid design. The core contribution of this work lies in the development of a dual-layered safety framework: a reactive layer that imposes BRA-based constraints only in critical configurations, and a proactive attentiveness module that dynamically assesses multi-agent risk and adjusts control objectives accordingly. Furthermore, the inclusion of a human-inspired disabling mechanism allows the system to avoid overly conservative behavior, bridging the gap between worst-case guarantees and realistic decision making. The proposed methodology is validated through simulations of different overtaking maneuvers involving multiple human-driven vehicles, both in same-lane and oncoming traffic settings. The results demonstrate the ability of the framework to balance safety, efficiency, and human-like behavior, achieving better safety performances than conventional minimum distance constraints or independent single-agent BRA constraints.

2. Hamilton-Jacobi reachability

This chapter focuses on Backward Reachability Analysis (BRA) as a powerful, but computationally demanding method for safety

verification. Reachability analysis determines whether a system can reach a specific set within a given time horizon. Given a system governed by $\dot{x} = f(x(t), u(t), d(t))$, where x is the state, u the control input, and d the disturbance, BRA propagates the system dynamics backward in time, identifying unsafe states that can lead to a target (unsafe) set. Compared to the open-loop approach of Forward Reachability Analysis (FRA), BRA provides robust safety guarantees under worst-case disturbances, reducing conservatism in a closed-loop framework. However, its computational complexity necessitates efficient numerical methods due to its reliance on dynamic programming. The Hamilton-Jacobi reachability framework offers a flexible approach for nonlinear systems by computing the Backward Reachable Set (BRS) via level set methods. The following sections formalize the definition of BRS, differentiate between sets and tubes and introduce Hamilton-Jacobi differential game theory.

2.1. System dynamics

The computation of the BRS allows for identifying all states that can lead the system into an unsafe configuration, under the presence of disturbances. This setting can be formulated as a two-player zero-sum differential game (Başar & Olsder, 1999). The inputs of both players are required to be measurable and bounded within the considered time interval. The system dynamics $f : \mathbb{R}^n \times \mathcal{U} \times \mathcal{D} \rightarrow \mathbb{R}^n$ is assumed to be uniformly continuous, bounded, and Lipschitz continuous (Chen & Tomlin, 2018a). Therefore, there exists a unique trajectory that solves the system dynamics, defined as:

$$\zeta(s; x, t, u(\cdot), d(\cdot)) : [t, 0] \rightarrow \mathbb{R}^n$$

It is important to specify the role and the information available to both players. For simplicity, the controlled vehicle (ego) is referred to as *player I*, while the adversarial disturbance vehicle is denoted as *player II*. The strategy of *player II* is defined as a non-anticipative mapping γ . Notably, *player II* can adapt its worst-case action at each time instant based on the control of *player I*, thus exploiting an informational advantage.

2.2. Backward reachable sets (BRS) and tubes (BRT)

The Backward Reachable Set (BRS) is defined as the set of states $x \in \mathbb{R}^n$ from which the system can be driven into a set $\mathcal{T} \subseteq \mathbb{R}^n$, called the *target set*, within a time horizon $|t|$. Depending on the problem formulation, two different definitions of BRS can be given (Chen & Tomlin, 2018b):

- The *maximal BRS* is defined as the maximal set of states from which the system can always be driven to the target set:

$$\mathcal{R}(t) = \{x \mid \forall \gamma \in \Gamma(t), \exists u(\cdot) \in \mathcal{U} : \zeta(0; x, t, u(\cdot), \gamma[u(\cdot)]) \in \mathcal{T}\}.$$

- The *minimal BRS* is defined as the set of states that will reach the target set for any control input:

$$\mathcal{A}(t) = \{x \mid \exists \gamma \in \Gamma(t), \forall u(\cdot) \in \mathcal{U} : \zeta(0; x, t, u(\cdot), \gamma[u(\cdot)]) \in \mathcal{T}\}.$$

If the target set $\mathcal{T}$ is interpreted as an avoid set, then the minimal BRS represents the set of states that will inevitably lead to unsafe configurations, regardless of the control strategy applied. Hence, safety is guaranteed as long as the system state remains outside the minimal BRS.

While the BRS defines reachability at the terminal time $|t|$, the Backward Reachable Tube (BRT) accounts for reachability over the entire time interval. This distinction is crucial in safety-critical scenarios where safety must be ensured along the entire horizon. The minimal BRT is formally defined as:

- *Minimal BRT*:

$$\bar{\mathcal{A}}(t) = \{x : \exists \gamma \in \Gamma(t), \forall u(\cdot) \in \mathcal{U}, \exists s \in [t, 0], \zeta(s; x, t, u(\cdot), \gamma[u(\cdot)]) \in \mathcal{T}\}.$$

2.3. Hamilton-Jacobi backward reachability

The Hamilton-Jacobi formulation casts the reachability problem as an OCP. In the two-player differential game setting, the cost functional is defined as:

$$J(x, u(\cdot), d(\cdot)) = \int_t^0 c(x, u(s), d(s)) ds + q(x(0))$$

where the time interval $[t, 0]$ is considered with $t < 0$. In the zero-sum game, *player I* seeks to maximize this cost, while *player II* aims to minimize it. Given the strategy of *player II*, the value function for the minimal BRS becomes:

$$V_{\mathcal{A}}(t, x) = \inf_{\gamma[u(\cdot)] \in \Gamma(t)} \sup_{u(\cdot) \in \mathcal{U}} J(x, u(\cdot), \gamma[u(\cdot)])$$

As shown in Evans and Souganidis (1984), the value function $V(t, x)$ is the viscosity solution to the Hamilton-Jacobi (HJ) partial differential equation:

$$\begin{cases} \frac{\partial V(t, x)}{\partial t} + H(x, \nabla V(t, x)) = 0, \\ V(0, x) = q(x) \end{cases} \quad (1)$$

where $H(x, \nabla V(t, x))$ is the Hamiltonian function, which depends on the system dynamics and the control formulation. For the minimal case, it is:

$$H_{\mathcal{A}}(x, \nabla V(t, x)) = \max_{u \in \mathcal{U}} \min_{d \in \mathcal{D}} [c(x, u, d, t) + \lambda^T f(x, u, d)]$$

with $\lambda = \nabla V(t, x)$ being the costate vector. Once the PDE is solved and the value function $V(t, x)$ is obtained, the optimal control for *player I* is:

$$u_{\mathcal{A}}^{Opt}(t, x) = \arg \max_{u \in \mathcal{U}} \min_{d \in \mathcal{D}} [c(x, u, d, t) + \lambda^T f(x, u, d)]$$

2.4. Solution of the backward reachable set problem

For computational tractability and practical understanding, the BRS computation is framed as a *game of degree* (Bansal et al., 2017) using the *level set method*, a set of algorithms that approximate the viscosity solution of the time-dependent PDE in Eq. (1) (Osher & Fedkiw, 2003).

The level set method requires the target set $\mathcal{T} \subseteq \mathbb{R}^n$ to be represented as the zero sublevel set of a bounded and Lipschitz continuous implicit function: $\mathcal{T} = \{x \in \mathbb{R}^n \mid g(x) \leq 0\}$, where $g(x)$ is the signed distance function to $\mathcal{T}$ border. Considering a cost functional with zero running cost and terminal cost $g(x)$, we have:

$$J(x, u(\cdot), d(\cdot)) = g(x(0)) \quad (2)$$

Assuming that $\mathcal{T}$ represents the unsafe set, the system reaches it if and only if $J(x, u(\cdot), d(\cdot)) < 0$. The resulting optimal control problem is formulated as a max-min game, starting from Eq. (1) and using the functional (2), the minimal BRS is the zero sublevel set of the viscosity solution $V(t, x) : \mathbb{R}^n \times [-T, 0] \rightarrow \mathbb{R}$, defined by:

$$\begin{cases} \frac{\partial V(t, x)}{\partial t} + \min [0, H(x, \nabla V(t, x))] = 0, \\ V(0, x) = g(x), \\ H(x, \nabla V(t, x)) = \max_{u \in \mathcal{U}} \min_{d \in \mathcal{D}} [\nabla V^T(t, x) f(x, u, d)], \\ \mathcal{G} = \{x \in \mathbb{R}^n \mid V(t, x) \leq 0\}. \end{cases} \quad (3)$$

To ensure safety over the entire time horizon, the BRT must be considered instead of the BRS. By augmenting the system, *player II* is assumed capable of *freezing* the dynamics once the trajectory of *player I* enters the target set. This is modeled by introducing the min operator in the PDE (3), implying that once a configuration is unsafe, it remains so throughout the horizon (Mitchell et al., 2005).

3. Backward reachability analysis on autonomous vehicle-human vehicle interaction

In this chapter, BRA is applied to the specific case of the human-autonomous vehicle interaction. The human-driven vehicle introduces uncertainty due to human behavior, which, while generally rational, can exhibit a certain degree of unpredictability. This scenario considers only one pair of autonomous and human-driven vehicles; however, in Chapter 4, the case involving one autonomous vehicle and multiple human-driven vehicles will be addressed.

Section 3.1 outlines the elements required to compute the BRT for the vehicle interaction problem, while Section 3.2 focuses on the visualization and interpretation of the BRT obtained in the overtaking scenario.

3.1. Backward reachable tube computation

The formulation of the reachability problem in the differential game setting outlines how to determine the minimal BRT by solving the HJ-PDE defined in Eq. (3). To solve this problem, three elements must be specified: the *system dynamics*, which describe the relative dynamics $\dot{x}_R = f_R(x_R, u, d)$ between the ego and the human vehicle; the *terminal set*, also known as the *avoid set*, including all unsafe configurations; the *Hamiltonian function* $H(x_R, \nabla V(t, x_R))$ and the *dissipation coefficients*.

As previously stated, reachability analysis suffers from the *curse of dimensionality*. Following the approach in Chen (2017), this work adopts a six-state kinematic model to describe the relative dynamics, balancing model accuracy and tractability. For the autonomous vehicle, a *five-state nonlinear single-track kinematic model* is used, relying on a key assumption: the front and rear wheel pairs are approximated as single wheels with zero slip angles. Consequently, the maximum speed of the autonomous vehicle is limited to 11 m/s. Another assumption is that motion is restricted to a planar surface, excluding pitch, roll, and vertical dynamics. The equations of motion of the five-state system in the global $X - Y$ coordinate frame are:

$$\begin{cases} \dot{X}_E = v_E \cos(\theta_E + \beta_E) \\ \dot{Y}_E = v_E \sin(\theta_E + \beta_E) \\ \dot{\theta}_E = \frac{v_E \cos(\beta_E)}{l_r + l_f} \tan \delta_E \\ \dot{\delta}_E = u_1 \\ \dot{v}_E = u_2 \end{cases} \quad (4)$$

where β_E is the vehicle slip angle defined as:

$$\beta_E = \arctan \left(\frac{l_r}{l_r + l_f} \tan \delta_E \right)$$

The human-driven vehicle is modeled as a *four-state dynamically extended Dubins car*. Its equations of motion in the global reference frame are:

$$\begin{cases} \dot{X}_H = v_H \cdot \cos(\theta_H) \\ \dot{Y}_H = v_H \cdot \sin(\theta_H) \\ \dot{\theta}_H = \omega_H \\ \dot{v}_H = a_H \end{cases} \quad (5)$$

Applying the same model of the ego vehicle to a human-driven one would require precise knowledge of the steering input. For this reason, the Dubins vehicle model is preferred for representing human agents, as the heading angle is more easily measurable via onboard sensors. Relative dynamics are defined with respect to a frame attached to the ego vehicle, with the X -axis aligned with its longitudinal direction. Thus,

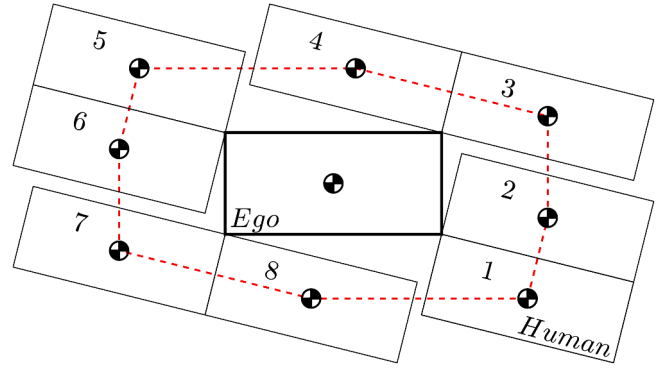


Fig. 1. Computation of the unsafe terminal set interface by fixing a relative heading angle (e.g. $\theta_{rel} = -14^\circ$).

the relative longitudinal and lateral positions x_{rel} , y_{rel} , and the relative heading θ_{rel} , are computed as:

$$\begin{bmatrix} x_{rel} \\ y_{rel} \\ \theta_{rel} \end{bmatrix} = \begin{bmatrix} \cos \theta_E & \sin \theta_E & 0 \\ -\sin \theta_E & \cos \theta_E & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} X_H - X_E \\ Y_H - Y_E \\ \theta_H - \theta_E \end{bmatrix}$$

Note that the same transformation cannot be applied directly to the velocities. The six-dimensional relative state vector $x_R \in \mathbb{R}^6$ is:

$$x_R = [x_{rel} \quad y_{rel} \quad \theta_{rel} \quad v_H \quad \delta_E \quad v_E]^T \quad (6)$$

and the corresponding relative dynamics are:

$$\begin{cases} \dot{x}_{rel} = v_H \cos \theta_{rel} - v_E \cos \beta_E + y_{rel} \omega_E, \\ \dot{y}_{rel} = v_H \sin \theta_{rel} - v_E \sin \beta_E - x_{rel} \omega_E, \\ \dot{\theta}_{rel} = \omega_H - \omega_E, \\ \dot{v}_H = a_H, \quad \dot{\delta}_E = u_1, \quad \dot{v}_E = u_2. \end{cases} \quad (7)$$

with:

$$\omega_E = \frac{v_E \cos(\beta_E)}{l_r + l_f} \tan \delta_E$$

The flow field f of the HJ-PDE represented by the relative dynamics between the autonomous vehicle and the human one is continuously differentiable over the considered states grid domain and over the admissible control inputs and disturbances. Thus Lipschitz continuity is satisfied, ensuring the existence and uniqueness of the solution of the HJ equation. Note that Lipschitz continuity would be lost only for steering angles of $\pm\pi/2$, which are unrealistic in any common vehicle application.

Determining the terminal set (TS) is essential to compute the BRT. To apply the level set method, the TS must be represented as the zero sublevel set of a Lipschitz continuous function, typically defined as the signed distance to the target set boundary. Assuming that collisions depend only on the relative position x_{rel} , y_{rel} , neglecting orientation and velocity, would lead to a circular approximation of the unsafe set centered on the ego vehicle's center of gravity. However, such an approach is overly conservative, as it ignores the actual vehicle geometries. A more accurate TS incorporates the effect of relative heading θ_{rel} , constructed via a rigid sliding process that ensures a continuous contact point between the ego and human vehicles, as shown in Fig. 1, repeated for each θ_{rel} .

The resulting TS is periodic in θ_{rel} with period π and offers a more realistic depiction of unsafe configurations. The ego vehicle dimensions are uniformly inflated with respect to the human vehicle ones, accounting for uncertainties in perception, control and modeling, and ensuring a conservative buffer in the reachability analysis.

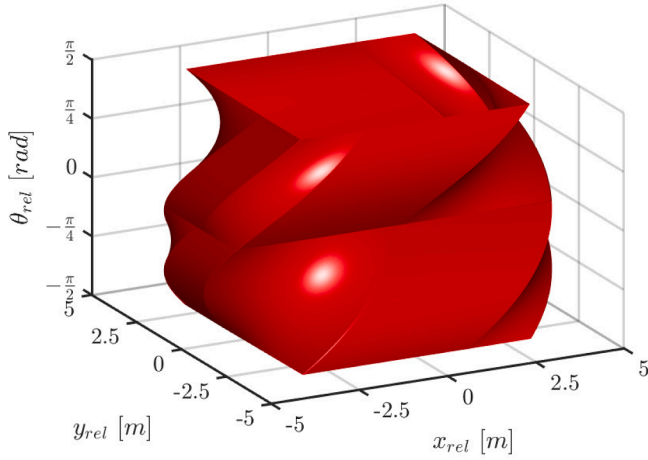


Fig. 2. Terminal set of all possible unsafe configurations, obtained considering the dimensions listed in Table 1 for the ego-human pair, for every relative angle configuration.

Table 1

Comparison of vehicle parameters for Ego and Human vehicles.

Vehicle parameters	Ego	Human
Length L [m]	4.68	4.28
Width W [m]	2.2	1.8
Front axle c.o.g. distance l_f [m]	0.959	0.959
Rear axle c.o.g. distance l_r [m]	1.706	1.706

To integrate this TS into the level set framework, its boundary is discretized over a grid, and a signed distance function is computed to represent $V(0, x_R)$, as the minimum distance between each grid point and the TS boundary, accounting also for the angular effect.

The BRT is computed using the Level Set Toolbox (Mitchell, 2004), which implements a discretized Hamiltonian based on the Lax-Friedrichs scheme:

$$\hat{H}(x_R, p^+, p^-) = H\left(x_R, \frac{p^+ + p^-}{2}\right) - \alpha^T(x_R) \left(\frac{p^+ - p^-}{2}\right) \quad (8)$$

where p^+, p^- are the spatial value function's gradient approximations. The second term is a dissipative contribution that stabilizes the numerical scheme by penalizing steep gradients (Osher & Fedkiw, 2003), where the dissipation coefficient is computed as:

$$\alpha_i = \max_p \left| \frac{\partial H(x_R, p)}{\partial p_i} \right|$$

The first term is the exact Hamiltonian for the two-vehicle interaction is:

$$H(x_R, p) = \max_{u_E \in U} \min_{u_H \in D} p^T \cdot f_R(x_R, u_E, u_H) \quad (9)$$

Using the relative control affine dynamics in Eq. (7), the Hamiltonian can be rewritten as:

$$H = f_x + \max_{\delta_E, v_E} (f_E) + \min_{\omega_H, a_H} (f_H)$$

where f_x contains only state-dependent terms, while f_E and f_H depend on the control inputs of the ego and human vehicles, respectively. The optimal control solution is the argument of the Hamiltonian resulting in a bang-bang strategy for Equation (3.1). Regarding the control limits, both agents are assumed to have the same allowable range for longitudinal acceleration, while the yaw rate ω_H of the human driver is modeled as velocity-dependent to avoid unrealistic behaviors. Lastly, input bounds are refined near grid edges to maintain numerical consistency.

Table 2

Limits for a high-resolution grid accounting for the human vehicle to be overtaken.

Params.	x_{rel} [m]	y_{rel} [m]	θ_{rel} [rad]	v_H [m/s]	δ_E [rad]	v_E [m/s]
Min.	-8	-6	$-\pi/4$	1	$-\pi/12$	1
Max.	17	6	$\pi/4$	11	$\pi/12$	11
Nodes	26	13	15	11	11	11

Table 3

Limits for a grid accounting for a distant oncoming human vehicle traveling in the opposite direction.

Params.	x_{rel} [m]	y_{rel} [m]	θ_{rel} [rad]	v_H [m/s]	δ_E [rad]	v_E [m/s]
Min.	-5	-12	$3\pi/4$	1	$-\pi/12$	1
Max.	70	12	$5\pi/4$	11	$\pi/12$	11
Nodes	26	13	15	11	11	11

Table 4

Limits for a grid accounting for a human vehicle traveling in the same direction coming from behind in the adjacent lane.

Params.	x_{rel} [m]	y_{rel} [m]	θ_{rel} [rad]	v_H [m/s]	δ_E [rad]	v_E [m/s]
Min.	-20	-6	$-\pi/4$	1	$-\pi/12$	1
Max.	5	6	$\pi/4$	11	$\pi/12$	11
Nodes	26	13	15	11	11	11

3.2. Backward reachable tube for realistic overtaking scenarios

The selected scenario involves an autonomous vehicle overtaking a human-driven vehicle either on a two-way road or on a two-lane highway. Consequently, only straight roads are considered as overtaking on a curve is not regulated by road code, while road curvature is negligible on highways, as stated in the related work paragraph, being consistent with the scenarios tested in the literature. Each ego-human couple is treated independently as a distinct pairwise interaction, following the approach of Wang et al. (2020), described by their relative dynamics and by the appropriate BRT. The formulation exploits the grid independence of the three BRT computations, which are solved separately. This is particularly advantageous given the limitations of dynamic programming on which the HJ-PDE solution is based on, which suffers from the curse of dimensionality. In realistic driving scenarios, many state combinations are never encountered, making it inefficient to compute a single high-dimensional BRT. Instead, decomposing the problem into smaller, independent subsets allows for more tractable computations. The use of separate grids ensures that each specific interaction is modeled with the precision it requires. The grid resolution must strike a balance between accuracy and computational feasibility. However, generating a full 6D grid with ideal resolution is impractical due to the high computational cost and memory usage. In particular, the relative heading range can be limited to $[-\pi/4, \pi/4]$ for vehicles traveling in the same direction and $[\pi - \pi/4, \pi + \pi/4]$ for those traveling in opposite direction. The x_{rel} domain is restricted in cases where the human vehicle does not significantly influence the ego vehicle. Three optimized grids are proposed:

For each grid, the human lateral input assumption is tuned empirically to yield meaningful results, with a steering capability reduction to 25%, 5%, and 25%, respectively for the three grids, with respect to the ego steering. Note that a greater steering capability assumption is unrealistic, as it would result in an unsafe setting whatever the relative configuration is during the overtake, meaning that this maneuver would always be considered unsafe to perform. These modifications improve computational efficiency while maintaining the accuracy of BRT-based safety evaluations. Note that the choice of the range of x_{rel}, y_{rel} is limited by the requirement of a closed implicit function by the level set method, avoiding numerical instabilities and invalid results. To maintain robustness, all grids are constructed to fully contain the initial terminal set.

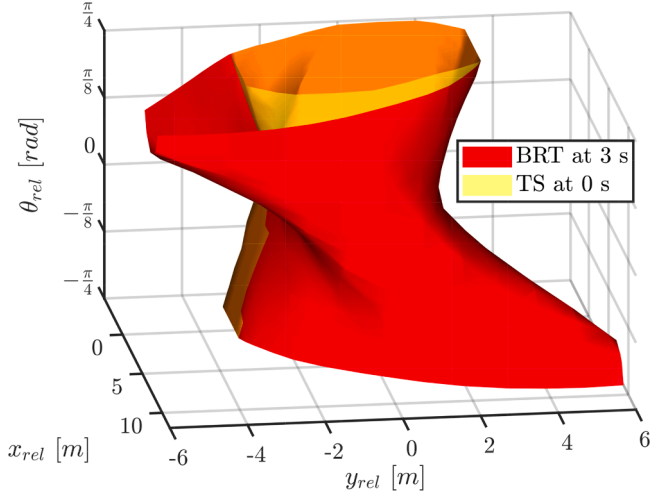


Fig. 3. 3D BRT visualization for fixed $v_H = 4 \text{ m/s}$, $\delta_E = 0 \text{ rad}$, $v_E = 10 \text{ m/s}$.

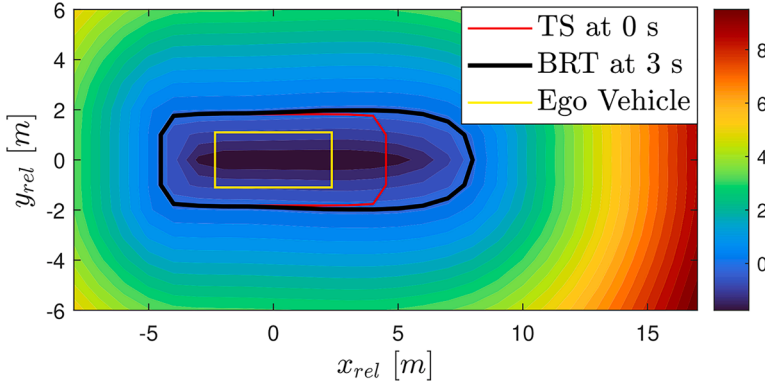


Fig. 4. 2D BRT visualization for fixed $\theta_{rel} = 0 \text{ rad}$, $v_H = 4 \text{ m/s}$, $\delta_E = 0 \text{ rad}$, $v_E = 10 \text{ m/s}$.

To further reduce computational burden, symmetries can be exploited. In particular, the system dynamics and the associated value function and its spatial gradients are symmetric in $y_{rel}, \theta_{rel}, \delta_E$, allowing the grid points to be halved:

$$V(x_{rel}, y_{rel}, \theta_{rel}, v_H, \delta_E, v_E) = V(x_{rel}, -y_{rel}, -\theta_{rel}, v_H, -\delta_E, v_E)$$

Since the BRT resides in a 6D space, full visualization is impractical. Instead, slices are taken by fixing three variables to produce interpretable 3D views. The examples shown refer to the BRT associated with Table 2, though the same methodology applies to the other cases. In this work, the BRT is computed over a 3 seconds horizon, which represents a practical trade-off, as it aligns with the MPC prediction horizon, later defined in 4, while avoiding overly conservative approximations that may arise in systems where the BRT does not converge. Fig. 3 shows a sample BRT propagation compared to the initial terminal set at $t = 0$.

Additional insights can be gained from 2D slices with fixed θ_{rel} , displayed using contour plots.

The value function, defined by Eq. (3), is used to assess the distance from the possibility of having an unavoidable collision, providing a rigorous, safety-related metric that indicates the potential for collision under worst-case assumptions. The value function goes beyond a simple Euclidean distance, as it captures the relative configuration of the system state. For instance, a higher relative velocity leads to a lower value, reflecting an increased risk of collision. This state-dependent formulation allows the value function to encode risk more precisely than a static metric. Furthermore, the Hamiltonian implicitly indicates how dangerous or safe the system will be, as it regulates the time evolution of the value function.

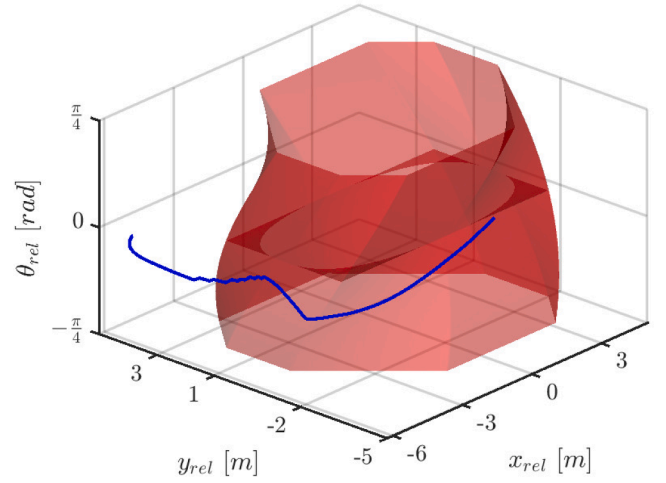


Fig. 5. Relative states trajectory between the human and autonomous vehicle in the $x_{rel}, y_{rel}, \theta_{rel}$ coordinate systems at 50 Hz.

To demonstrate the computed BRT, a pursuit-evasion game is simulated in MATLAB & Simulink, involving an *evader* (ego vehicle) and a *pursuer* (human vehicle). Starting from a safe state where $V(t, x_R) > V_{thr}$, the ego initially performs a trajectory tracking control while the pursuer applies optimal disturbances. When V drops below V_{thr} , an optimal control strategy is triggered to preserve V from decreasing, the ego switches to an avoidance mode, successfully preventing the entry into the unsafe set within a prediction horizon of $|T| = 3 \text{ s}$ (Fig. 5).

4. MPC motion planner in overtaking scenario with backward reachability analysis

This chapter presents the implementation of a Model Predictive Control (MPC) framework to compute the optimal control strategy in a multi-agent overtaking scenario. The optimal control problem (OCP) formulation includes:

- the vehicle model along with state, input, and safety-preserving constraints;
- the cost functional;
- the control strategy for overtaking in multi-agent scenarios.

4.1. Vehicle model and problem constraints

The vehicle model used in this work is a five-state nonlinear single-track kinematic model expressed in global $X-Y$ coordinates, as reported in Eq. (7). State, input, and path constraints are defined to ensure feasible and safe trajectories. Hard constraints include road boundaries and physical limits on steering and velocity, while soft constraints penalize deviations from reference lane positions and low speeds. The road boundaries are defined as $y_{left} = 4 \text{ m}$ and $y_{right} = -4 \text{ m}$. Control variables are bounded due to actuator saturation, with steering rate $\dot{\delta}$ and acceleration $\dot{v}$ limits consistent with those used in the BRT computation.

Following the approach of Leung et al. (2020), the BRT is integrated into the MPC planner as the first safety-preserving layer of the framework. When approaching near unsafe configurations, rather than switching to an optimal collision-avoidance control, the vehicle selects inputs that minimally deviate from the optimal trajectory, maintaining both safety and goal achievement. Control actions must belong to the set of inputs that do not decrease the value function, defined as:

$$\mathcal{U}_{safe}(x_R) = \left\{ u \mid \min_{u_H} (\nabla V^T(t, x_R) \cdot f_R(x_R, u, u_H) \geq 0) \right\} \quad (10)$$

This constraint is enforced for every human vehicle when $V(x_R) \leq V_{thr}$, with threshold values selected based on the simulation frequency. Specifically:

- $V_{thr}^{(j)} = 0.7$ for the vehicle to be overtaken;
- $V_{thr}^{(j)} = 1$ for vehicles in adjacent lanes.

where j denoted the j -th human vehicle. The BRT-based constraint, referred to as the reactive safety layer and triggered with a predefined safety margin, ensures that the system never reaches a state where a collision is physically possible, corresponding to $V < 0$. The activation thresholds for the BRT constraints, considering the definition of the terminal set, the resolution of the grid, and the input, are deliberately conservative also to account for numerical errors and modeling uncertainties, as well as for possible discretization inaccuracies in the BRT caused by grid sampling thereby enhancing the overall robustness of the safety layer. Since the Hamiltonian is control affine, the constraint (10) can be linearized as:

$$\tilde{U}_{safe}(x_R) = \{u \mid M_{HJ}^T \cdot u + b_{HJ} \geq 0\} \quad (11)$$

To guarantee feasibility, a slack variable σ_{HJ} is introduced, allowing controlled constraint violation, penalized in the cost function:

$$\begin{cases} M_{HJ}^T \cdot u + b_{HJ} \geq -\sigma_{HJ}^{(j)} \\ \sigma_{HJ}^{(j)} \geq 0 \end{cases}$$

Additionally, a soft constraint based on geometric distance is introduced to assess the effectiveness of the reachability constraint, as detailed in Chapter 5. The predicted position of the human vehicle is computed under a constant velocity assumption. Both vehicles are approximated as circles to reduce computational load. The distance-based constraint is defined as:

$$C_{dist,n,q}^{(j)} = e^{-\left[(x_{E_n} - x_{H_q}^{(j)})^2 + (y_{E_n} - y_{H_q}^{(j)})^2 - (\rho_E + \rho_H)^2\right]} \quad (12)$$

where n and q denote two circles used to approximate the shape of the vehicle and are used to define the distance between the two vehicles. To compare the proposed approach with a baseline method, the constraint C_{dist} is formulated exclusively for comparison purposes, in order to highlight the differences between our method, which preserves a positive value of the value function to ensure a safe trajectory, with respect to a simple baseline method which relies on preserving the euclidean distance from another vehicle.

Moreover, to discourage unsafe right side overtaking, a penalty is applied for each human-driven vehicle m to be overtaken:

$$C_{overtake}^{(m)} = e^{-\left[y - y_H^{(m)} + k(x - x_H^{(m)})^2\right]} \quad (13)$$

4.2. Human-like autonomous vehicle overtake control strategy with reachability danger assessment

The MPC framework discussed so far efficiently handles overtaking scenarios involving only the ego and a single human-driven vehicle. However, when multiple human vehicles are present, the current safety-preserving reachability constraints and other associated constraints do not account for mutual interactions between the ego and multiple external agents. This can result in conflicting optimal safety-preserving controls, potentially leading to unsafe configurations. Therefore, a control strategy capable of managing multiple safety-preserving constraints in multi-agent scenarios is developed.

Instead of relying on a fixed value function threshold, as usually done in literature leading to overly conservative behavior, a dynamic threshold $V_{dyn}(t, x_R)$ is introduced to better emulate human drivers' dynamic risk assessment. This threshold depends on relative vehicle speeds and distances, allowing for more realistic decision-making in multi-agent

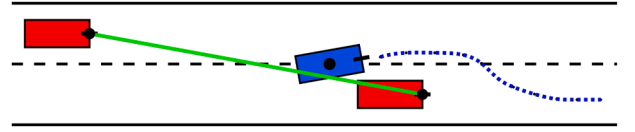


Fig. 6. Emergency merge reference maneuver.

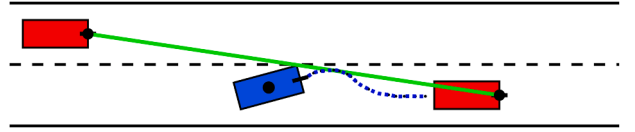


Fig. 7. Emergency follow reference maneuver.

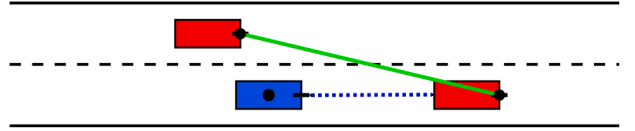


Fig. 8. Wait reference maneuver.

contexts. The attentive level is triggered when two or more human vehicles exceed this dynamic threshold, defined as:

$$V_{dyn,j}(x_R) = V_{thr,j} + k_{vel} v_H(t) + |v_E(t) - v_H(t) \operatorname{sgn}(\cos \theta_{rel}(t))| (y_{rel}(t) > 0) \quad (14)$$

where k_{vel} is a weighting coefficient. When the attentive level is triggered during an overtaking maneuver, the autonomous vehicle evaluates the situation and select an appropriate fallback strategy. Traditional reachability analysis could fail in preserving safety when applied to multiple interactions simultaneously due to the presence of several BRT constraints. This motivates the introduction of a proactive mechanism as the attentive level, which allows the system to adjust its behavior before reaching configurations that would be difficult to manage. In this way, there is no conflict between multiple BRT constraints because the dynamic threshold anticipates potential risk and shifts the ego's reference to maintain a manageable and safe configuration. Beyond the nominal tracking control, three distinct control modes are further defined when this level is triggered, based on the relative positions of nearby human-driven vehicles:

- **Emergency merge:** The vehicle accelerates and completes the overtake by merging ahead of the vehicle being overtaken. This strategy is used when completing the maneuver is safer than returning to the original lane to nearby traffic.
- **Emergency follow:** The vehicle aborts the overtaking attempt and falls back behind the vehicle it intended to overtake. This option is selected when completing the maneuver would be unsafe due to surrounding or oncoming traffic.
- **Wait:** When no immediate threat is detected and neither one of the previous strategies is required, the vehicle enters a waiting mode, queuing behind the current traffic until a safe overtaking opportunity becomes available.

While minimal BRA effectively captures worst-case scenarios, it lacks the flexibility and intuitive judgment that human drivers apply in complex traffic situations. To improve maneuver smoothness and reduce conservatism, it is crucial to disregard configurations that a human driver would not perceive as dangerous. Introducing human-like rational behavior helps overcome some limitations of reachability analysis.

Consider the case of two vehicles traveling straight in parallel lanes. The reachability analysis may classify this as extremely dangerous because, in the worst-case, a sudden steering maneuver could cause an unavoidable collision. However, human drivers rely on social conventions

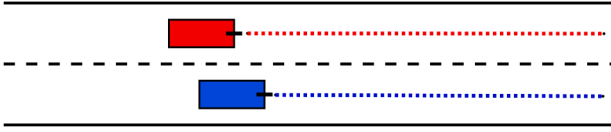


Fig. 9. Human-like trajectory triggered by condition (15).

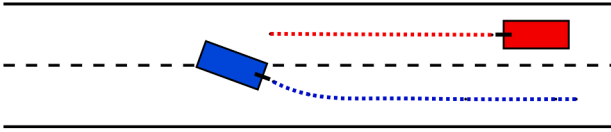


Fig. 10. Human-like trajectory triggered by condition (16).

and implicit behavioral understanding factors not captured by reachability alone. Thus, a balance is sought between worst-case safety guarantees and a realistic interpretation of human intent. For ego-human configurations showed in 9 the safety-preserving constraint is disabled even when the value function drops below the threshold:

$$|y - y_H^{(j)}| - W_{lane} < \varepsilon_y \quad \text{and} \quad \text{mod}(|\theta|, \pi) < \varepsilon_\theta \quad (15)$$

where W_{lane} denotes the width of a single lane while ε_y and ε_θ are small threshold values.

The constraint is also disabled for ego-human configurations as showed in 10. Such conditions are identified experimentally and excluded from the constraint:

$$y_{rel} < 0 \quad \text{and} \quad \text{mod}(|\theta|, \pi) > \varepsilon_\theta \quad \text{and} \quad y_H^{(j)} > 0 \quad (16)$$

The proactive module activates in the presence of two or more agents and adding an additional vehicle would simply make the ego more cautious with respect to the surrounding configuration. In fact, considering just two human vehicles is often sufficient to capture the vast majority of relevant interactions.

4.3. MPC: Overall formulation

Summarizing the model, constraints, and control logic described in this chapter, the overall formulation of the Model Predictive Control (MPC) framework is expressed as:

$$\begin{aligned} \min_{X, U, S} \quad & \frac{1}{2} \int_{t_0}^{t_f} \left\{ [X(\tau) - X_{ref}(\tau)]^T Q [X(\tau) - X_{ref}(\tau)] + \right. \\ & [U(\tau) - U_{ref}(\tau)]^T R [U(\tau) - U_{ref}(\tau)] + \\ & S(\tau)^T Z S(\tau) \Big\} d\tau + \\ & \frac{1}{2} [X(t_f) - X_{ref}(t_f)]^T P [X(t_f) - X_{ref}(t_f)] \end{aligned}$$

subject to :

$$\begin{aligned} \dot{x}(t) - f(x(t), u(t)) &= 0 \quad \forall t \in [t_0, t_f] \\ x(0) - x_0 &= 0 \\ \delta_{\min} \leq \delta(t) \leq \delta_{\max} &\quad \forall t \in [t_0, t_f] \\ \dot{\delta}_{\min} \leq \dot{\delta}(t) \leq \dot{\delta}_{\max} &\quad \forall t \in [t_0, t_f] \\ v_{\min} \leq v(t) \leq v_{\max} &\quad \forall t \in [t_0, t_f] \\ \dot{v}_{\min} \leq \dot{v}(t) \leq \dot{v}_{\max} &\quad \forall t \in [t_0, t_f] \\ D \sin(\alpha + |\theta(t)|) + y(t) - y_{\text{left}} &\leq 0 \quad \forall t \in [t_0, t_f] \\ D \sin(\alpha + |\theta(t)|) - y(t) + y_{\text{right}} &\leq 0 \quad \forall t \in [t_0, t_f] \\ \sigma_{HJ}^{(j)} \geq 0 \quad \text{if} \quad V^{(j)}(t, x_{\text{rel}}^{(j)}) &\leq V_{thr}^{(j)} \\ M_{HJ}^{T(j)} \cdot u + b_{HJ}^{(j)} \geq -\sigma_{HJ}^{(j)} \quad \text{if} \quad V^{(j)}(t, x_{\text{rel}}^{(j)}) &\leq V_{thr}^{(j)} \\ C_{overtake}^{(m)} &\geq 0 \end{aligned} \quad (17)$$

where $t_f = t_0 + 3$ s, while D and α are dimensional parameters. Q , R , P , Z are the weighting matrices for the states, inputs, terminal state, and slack variables, respectively. These weights are selected to achieve a trade-off between trajectory tracking accuracy, control effort and safety enforcement. The slack variables vector S incorporate penalties associated with safety constraints and overtaking constraints.

The dynamics and initial condition constraints are included to ensure proper formulation of the OCP. It is essential to emphasize that the value function and the cost function are fundamentally different objects. The cost function is explicitly defined in the MPC formulation to encode performance objectives. In contrast, the value function arises from a differential game formulation and quantifies safety through BRA. The reference trajectories X_{ref} and U_{ref} correspond to nominal straight driving behavior, unless the attentiveness level is triggered setting as reference one of the fallback strategies: *emergency merge*, *emergency follow*, or *wait*.

Moreover, for comparison purposes, in Chapter 5, the geometrical distance constraint of Eq. (12) is enforced only when explicitly stated.

5. Simulations and results

This chapter presents the simulation environment used to validate the proposed MPC framework and the safety assessment based on BRA, implemented in MATLAB with *acados*. First, validation is performed in a simple overtaking scenario involving one human-driven vehicle, to test the correctness of the problem formulation and constraints. Subsequently, a second human vehicle is introduced to evaluate the attentive and reaction levels in a multi-agent setting, considering both same-direction and opposite-direction traffic configurations.

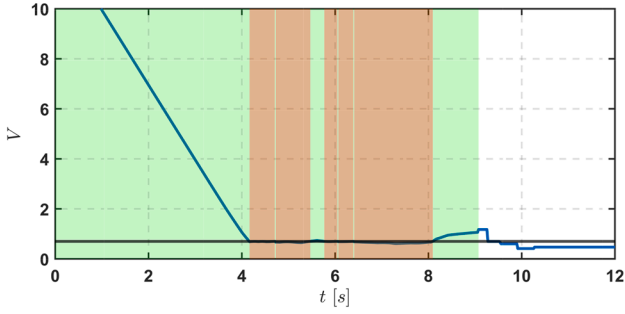
Simulation is the safest and most scalable option for evaluating such interactions, particularly when studying the effect of constraint activation or violation in worst-case conditions.

All simulations are performed on a two-lane road in compliance with road regulations. A simulation is considered failed if the ego vehicle collides with the road boundaries or if the value function between the ego and any human vehicle drops below zero, excluding cases where the common sense disabling conditions apply.

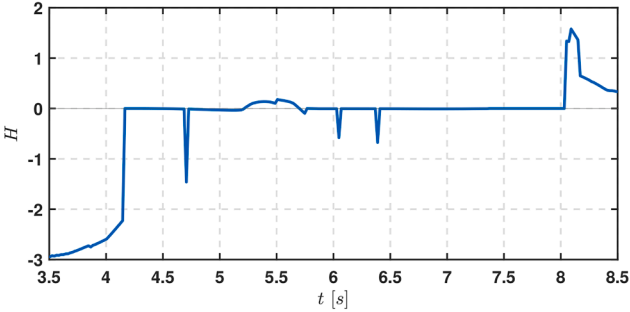
The simulation framework continuously updates the ego vehicle's states and interpolates the relative states with respect to each human vehicle into the corresponding BRT. Safety-preserving reachability constraints are then triggered according to the value function thresholds.

Notice that the goal of this work is not to estimate the aggressiveness or risk level of the human driver, but rather to construct a robust safety framework with a realistic rational behavior. This implies that the human-driven vehicle is assumed to behave rationally, avoiding clearly unsafe actions that would lead to inevitable collisions such as intentionally driving into the ego vehicle. This assumption strikes a balance as it rules out unrealistic behavior while preserving enough uncertainty to keep the interaction challenging. However, the human vehicle is modeled with a small white noise in its inputs to add a layer of realism. Nonetheless, the proposed approach could be extended by incorporating estimation techniques to infer driver behavior, allowing the BRT to adapt accordingly. In such a case, an additional state could be introduced to model the human driver's ability to execute certain inputs, influencing the reachability computation. The paper also investigates the effect of increasing the input bounds, showing how the value function adapts to more aggressive behaviors. Note that assuming constant speed makes the analysis more conservative ensuring that safety is preserved even in the absence of cooperative behavior.

The MPC is solved in MATLAB using *acados* (Verschuere et al., 2021) and executed at 50 Hz with a prediction horizon of 3 s, balancing computational efficiency and solution accuracy to meet the real-time requirements of the framework. The solver configuration in *acados* uses the Sequential Quadratic Programming (SQP) scheme combined with the High-Performance Interior Point Method (HPIPM) quadratic program solver with partial condensing, with warm-starting for both the QP solver and the first SQP iteration to improve performance in



(a) Value function of the ego vehicle with respect to the human car.



(b) Hamiltonian time evolution.

Fig. 11. Value function and Hamiltonian visualization.

real-time applications. This setup provides a good trade-off between computational efficiency and solution accuracy in nonlinear MPC. Note that all subsequent simulations share the same constraint formulations and control parameters, differing only in the initial conditions of the vehicles, thereby illustrating the generality and adaptability of the proposed approach. The simulations were conducted on a notebook equipped with an Intel Core i9-13980HX CPU, 32 GB DDR5 RAM, and a NVIDIA RTX 4060 GPU.

5.1. Overtake of one human vehicle

In this simulation, the ego vehicle attempts to overtake a slower-moving human-driven vehicle. As it approaches the human vehicle, the ego deviates from its lateral reference trajectory due to the imposed overtaking penalty cost and the activation of the safety-preserving reachability constraint. The overtaking maneuver is considered completed once the ego vehicle is sufficiently ahead of the human vehicle.

The evolution of the value function is shown in Fig. 11a, illustrating the ego vehicle's proximity to the unsafe set. When the value function falls below a predefined threshold, the safety-preserving constraint is triggered, leading to stabilization of the value function, as depicted in Fig. 11b.

(For all value function visualizations, the following color convention is adopted: green background indicates that the relative states lie within the corresponding BRT grid; red background denotes that the safety-preserving constraint is active; white background signifies that the relative states are outside the BRT grid and the interaction is over.)

For a clear visualization of the overtake maneuver, refer to Fig. 12.

As the human steering authority increases, the ego vehicle follows a wider trajectory (Fig. 13), illustrating the impact of more conservative BRT assumptions and the importance of a realistic calibration of human input.

To assess the effectiveness of the reachability analysis, the same simulation is repeated, replacing the safety preserving constraint with the geometrical one in (12), to give the ego information about distant ve-

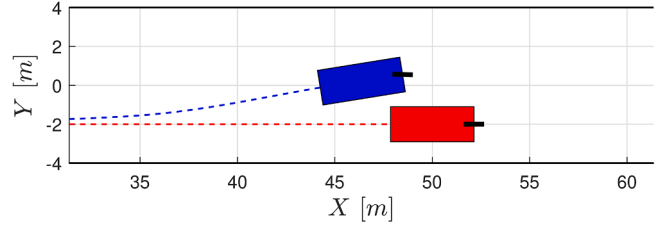
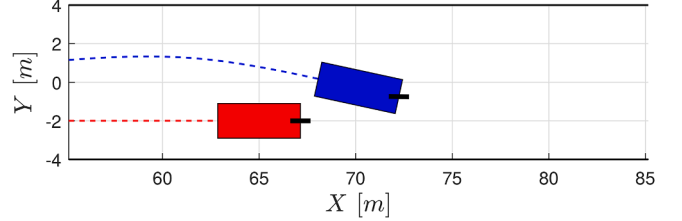
(a) Snapshot at $t = 6$ s. Safety-preserving constraint is active.

Fig. 12. Relevant snapshots related to the overtake of one human vehicle (Blue: ego vehicle. Red: human vehicle).

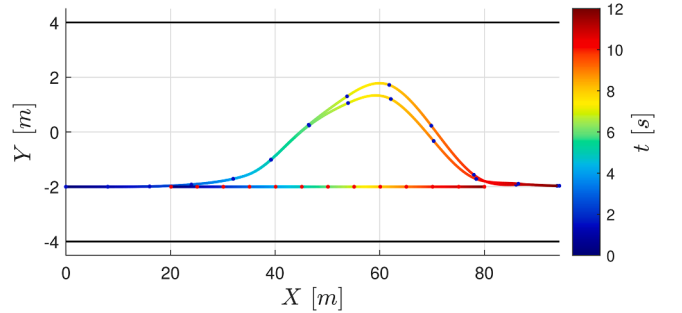


Fig. 13. Comparison of time evolution of the trajectories for different human steering assumptions in the adopted BRT. The wider trajectory corresponds to a reduction of 50%, the smaller of 25%.

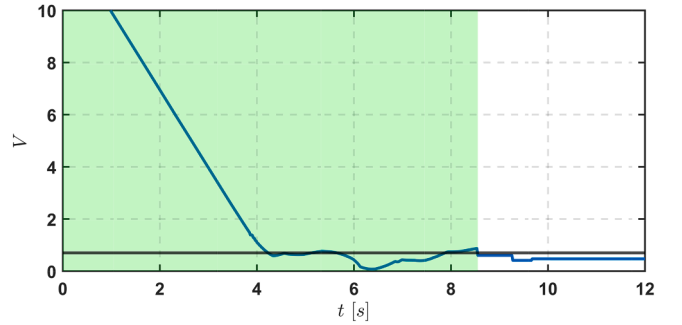


Fig. 14. Value function of the ego vehicle switching off the safety-preserving constraint and replacing it with the geometrical distance constraint only, resulting in an unsafe configuration.

hicles in place of the reachability constraint. As shown in Fig. 14, this leads to a dangerous interaction, where the value function drops to zero, signaling a potential unsafe configuration.

5.2. Overtake with a second human vehicle coming in the opposite direction

This section expands the analysis to a more complex scenario, where the ego vehicle must navigate around a slower human vehicle in the same lane while also accounting for a second human vehicle traveling in the opposite direction. The initial conditions of the ego and H_1 (the human vehicle in the same lane) remain the same as in

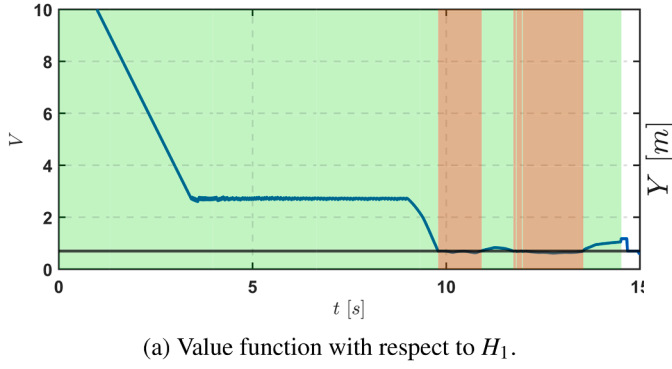


Fig. 15. Value functions of the ego vehicle during an overtake with a second human vehicle coming in the opposite direction.

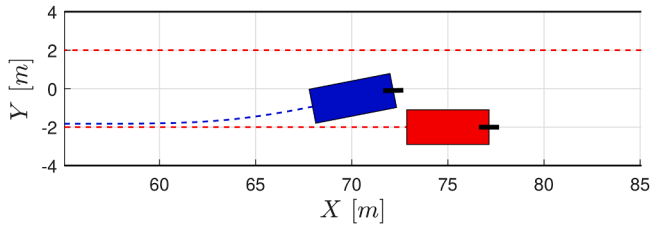
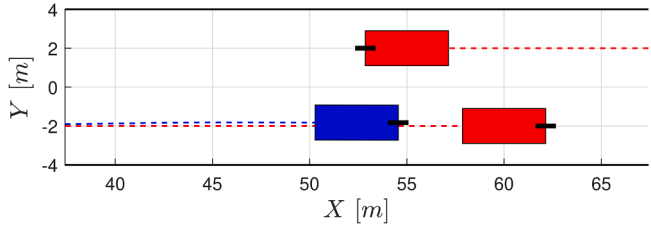


Fig. 16. Relevant snapshots related to the vehicles trajectories of 17.

Section 5.1, ensuring consistency with the single-agent scenario. In contrast, the initial conditions of H_2 (the second human vehicle) are deliberately chosen to evaluate the ego's response in a more complex, multi-agent setting. The grids used for reachability analysis are defined as in Tables 2 and 3 for H_1 and H_2 , respectively. Specifically, the positioning of H_2 is set to trigger the *wait* control mode in the autonomous vehicle, allowing analysis of its ability to temporarily defer overtaking when conditions are not favorable.

The goal is to assess whether the ego vehicle recognizes the danger posed by H_2 and waits until performing the overtake over H_1 is safe. Initially, the ego vehicle detects an oncoming vehicle in the opposite lane, preventing an overtaking maneuver. It decelerates and waits for H_2 to pass before completing the maneuver due to the activation of the attentive level in eq.(14), that causes the switch in the ego vehicle reference enforcing the wait condition. The ego maintains a constant distance from H_1 between 3 and 9 seconds resulting in a constant value function with respect to H_1 , even though the reactive safety-preserving constraint is not yet enforced in the OCP.

As the attentive level is activated, the autonomous vehicle changes its reference according to the wait condition.

The ego vehicle slows down to queue behind H_1 (Fig. 16a) until the overtake is safe, i.e., H_2 has completely passed the ego (Fig. 16b). The complete trajectory, including all the three vehicles, is shown in Fig. 17.

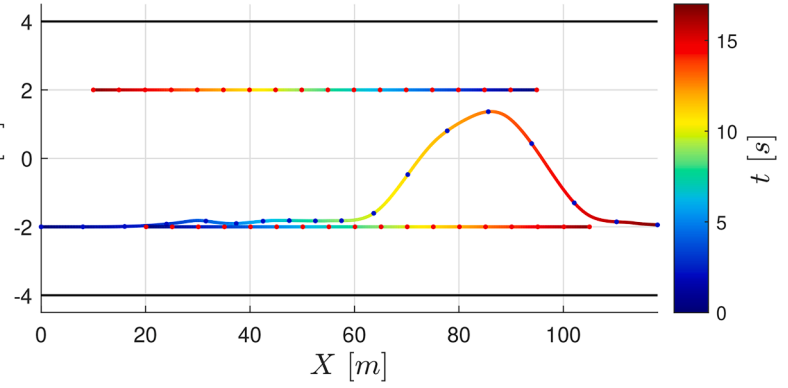
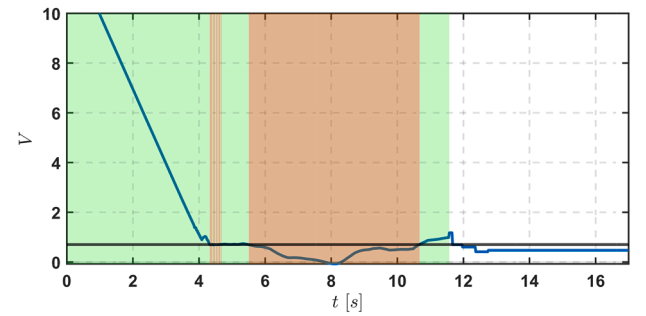


Fig. 17. Vehicle trajectories and speed profiles with the *wait* reference. The ego slows down to follow H_1 .



(a) V_1 with geometrical distance and two independent reachability constraints ($V_1 < 0$ at 8s).

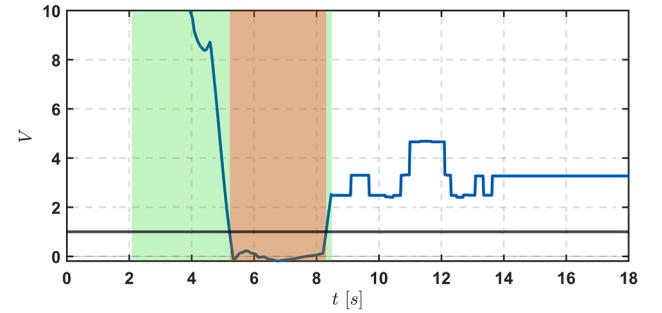


Fig. 18. Safety violations due to removal of the attentive level module under different constraint settings.

Replacing the attentive level module with the previously defined distance geometric constraint, the ego vehicle's reference does not change. In this case, the vehicle fails to react appropriately to the danger posed by H_2 , resulting in safety violations ($V < 0$), as shown in Fig. 18.

In particular, attempting to enforce two independent reachability constraints leads the ego vehicle into an unsafe configuration, as illustrated in Fig. 19. If H_2 's longitudinal initial condition is changed, it affects the autonomous vehicle's response. With a farther initial position for H_2 , the ego vehicle delays its detection of the danger posed by H_2 and begins the overtaking maneuver.

Upon recognizing the threat, the attentive level is activated, and the ego vehicle switches to the *emergency follow* reference, decelerating until it is safe to complete the overtake while preserving safety with respect to both value functions. As before, a greater human steering

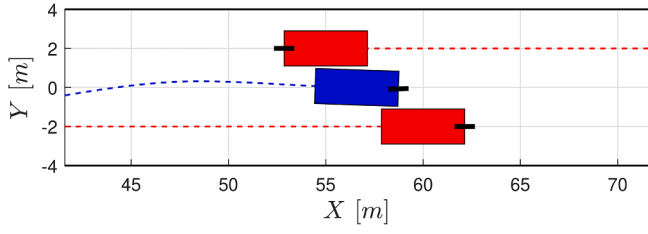
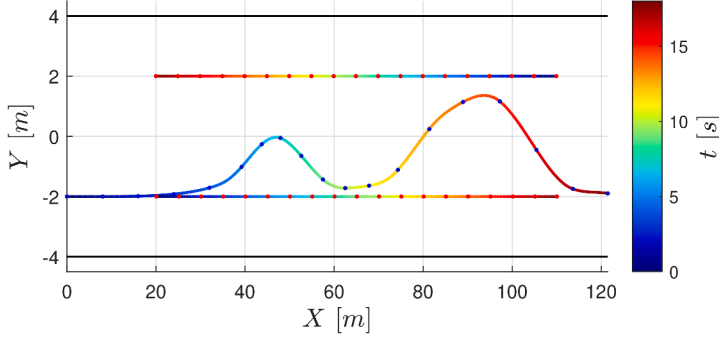
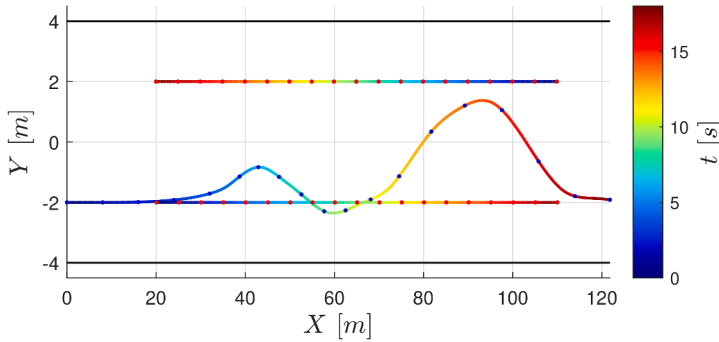


Fig. 19. Achieved unsafe configuration without enforcing the attentive level.



(a) Time trajectories of the vehicles.



(b) Time trajectories of the vehicles with increased human input assumption.

Fig. 20. Emergency follow application for different human lateral input assumptions.

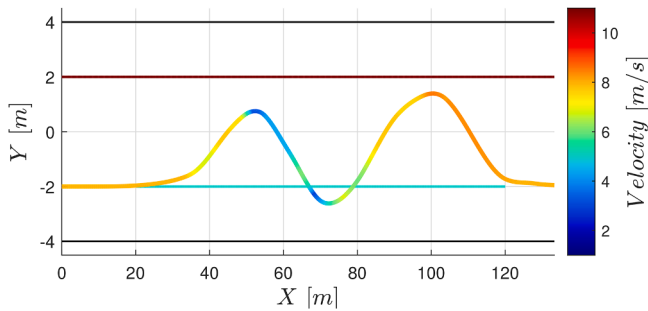


Fig. 21. Vehicle trajectories and speed profiles with the emergency follow reference with high speed H_2 .

capability assumption on the lateral input ω_H for H_2 leads to a more abrupt response by the ego vehicle, as shown in Fig. 20.

Additionally, if the velocity of H_2 is increased (and its longitudinal position adjusted) the ego must perform an emergency maneuver.

This leads to a first deceleration to abort the overtake and a second one to avoid impact with road boundaries, as shown in Fig. 21.

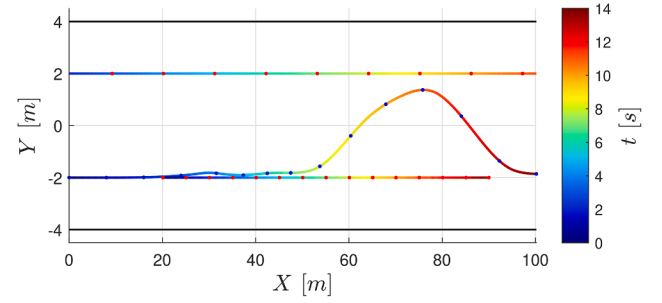


Fig. 22. Visualization of vehicles trajectory time evolution applying the wait reference.

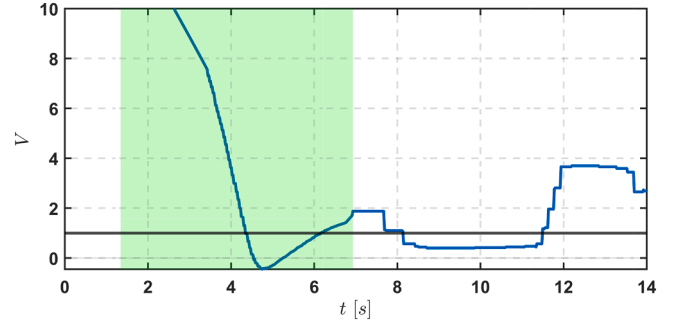


Fig. 23. Value function with respect to H_2 in case of common sense activation.

5.3. Overtake with a second human vehicle coming in the same direction

This section analyzes the case in which a second human-driven vehicle approaches from behind in the same direction, with the rest of the simulation environment remaining unchanged. The initial conditions of the ego and the overtaken vehicle are the same as in the previous scenario, while the initial conditions of the second human vehicle are selected to interact with the ego vehicle during the overtake and to trigger different attentive level control modes. The grid for H_1 is defined in Table 2, whereas the BRA for H_2 is conducted using the grid defined in Table 4. Initially, H_2 is positioned so that it engages the ego vehicle before the overtaking maneuver begins, triggering the wait condition. Consequently, the ego vehicle queues behind H_1 until H_2 overtakes both vehicles, as illustrated in Fig. 22.

Although the value function with respect to H_2 drops to zero, as in Fig. 23, implying that a collision would be inevitable from a reachability perspective, the scenario is still treated as safe because it relies on the realistic assumption that a human driver would never attempt a steering maneuver in such conditions, as previously reported in Fig. 9.

The common sense infusion, described in Equation (15), explains that although the value function approaches zero, the configuration is safe, as no human driver would ever swerve into the ego when completely flanked. If the common sense module is disabled, the ego exhibits unrealistic behavior, as shown in Fig. 24, preserving the value function with H_2 above its threshold.

If H_2 's longitudinal initial position is chosen farther from the autonomous vehicle, the ego vehicle begins the overtake with no vehicles detected beyond H_1 . Upon detecting H_2 , the ego triggers the emergency merge maneuver, prioritizing completing the overtake with maximum acceleration for safety, as shown in Fig. 25.

The simulation results, shown in Fig. 26, confirm that the ego increases speed to avoid H_2 and to force the completion of the overtake maneuver.

The scenario is treated as safe since it relies on the common-sense assumption that a human driver would never attempt a steering maneuver in such conditions but it would wait before completing the overtake.

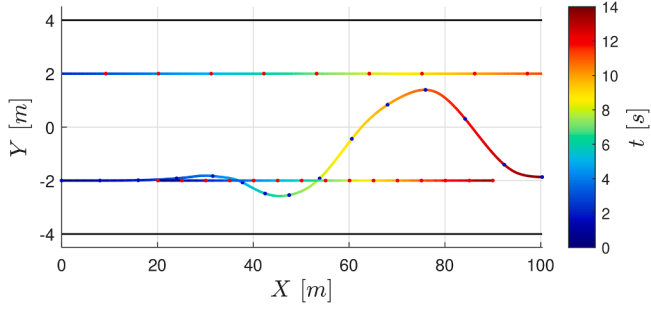


Fig. 24. Trajectory of the vehicles over time without enforcing the common sense. The ego unnecessarily swerves.

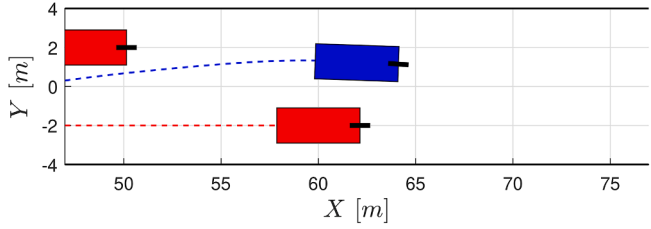


Fig. 25. Snapshot of the triggering configuration of the emergency merge maneuver.

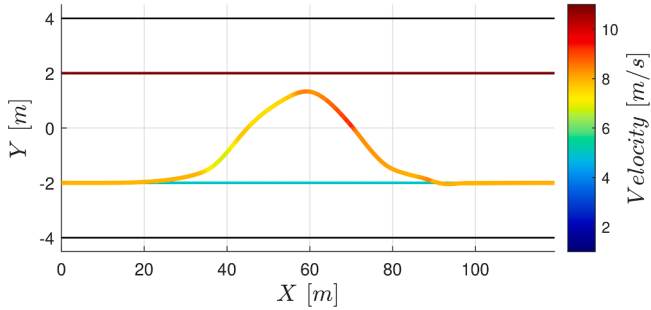


Fig. 26. Visualization of vehicles trajectory evolution and velocities profile applying the emergency merge reference. The ego accelerates to safely complete the overtake.

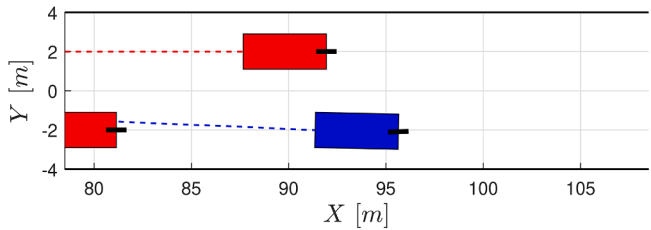


Fig. 27. Snapshot of the triggering configuration of the common sense infusion of Eq. 16.

Thus, the disabling module is triggered when the vehicles are almost flanked, as depicted in Fig. 9.

If the disabling module is not enforced, the ego would unnecessarily swerve to avoid H_2 , as shown in Fig. 28, in order to preserve safety from a reachability point of view, thus demonstrating the practical relevance of the proposed framework.

To validate the efficiency of the framework, the execution time of an MPC run, accounting for different number of human-driven vehicles, is reported in Fig. 29. Note that multi-agent safety is enforced when more than one human vehicle is present, thus, increasing the number

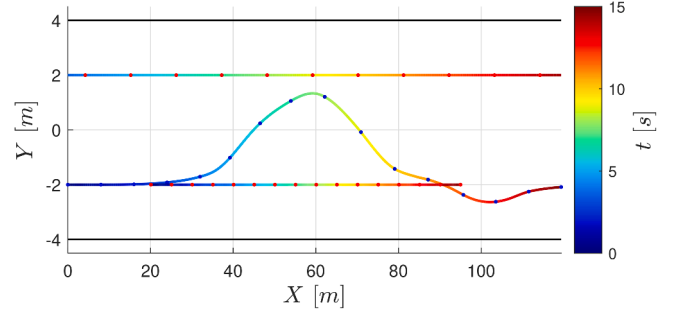


Fig. 28. Trajectory of the vehicles over time without enforcing the disabling module. The ego unnecessarily swerves.

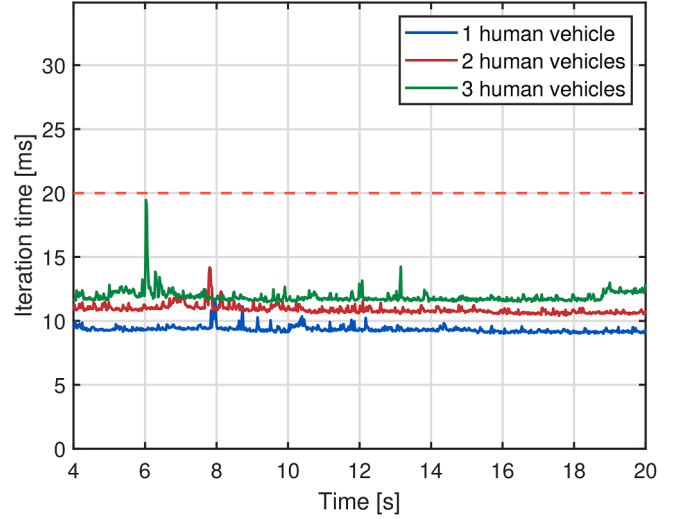


Fig. 29. Solver computational time per iteration for different number of human-driven vehicles.

of human vehicles would just trigger earlier the activation of the multi-agent safety layer.

The full implementation of the proposed control framework is available at the following link: [ARGOLab-Polimi-Reachability](https://argolab.polimi.it/reachability).

In addition, a series of videos illustrating each of the scenarios discussed in this section is provided to offer further insight into the controller's behavior and performance.

6. Conclusion and future developments

This work investigates the integration of BRA within an MPC framework for autonomous vehicles operating in realistic multi-agent scenarios. The focus is set on safely executing autonomous overtakes while considering multiple human-driven vehicles. Since human behavior is uncertain, BRA assumes bounded rationality to compute the BRT by solving the Hamilton-Jacobi PDE via the Level Set Method, identifying all ego-human relative configurations that could lead to a collision despite optimal control. To address the curse of dimensionality, three decoupled BRTs are used, leveraging grid symmetry and scenario-based gridding. Accuracy and tractability are verified through grid analyses and pursuit-evasion simulations.

Based on these results, an MPC strategy is developed with nonlinear constraints that incorporate safety-preserving constraints. The ego vehicle aims to maintain lane position and longitudinal progression, overtake when necessary, while remaining outside unsafe sets for each ego-human pair. Moreover, a novel safety layer in the decision-making module, which, depending on a dynamic threshold that depends on the relative configuration between the ego and each human vehicle, is added

to assess risk in multi-agent scenarios. Additionally, a common sense layer is implemented to mimic human driving rationality by disabling the BRT constraint when a safe configuration is mistakenly labeled as dangerous. Validation in two-lane scenarios confirms the ego vehicle safely overtakes slower traffic and avoids oncoming vehicles, showing better performances than baseline geometric constraints and than independent BRA methods. Moreover, the MPC execution time remains sufficiently low while ensuring high frequency real-time applicability.

This work addresses the computational cost, single agent focus, and over conservatism highlighted by the current state of the art, delivering a comprehensive risk-aware BRA framework for safe and efficient autonomous overtaking in multi-agent scenarios.

Future research directions include the extension of the reachability formulation by introducing additional states to capture variations in human input, making the approach even more flexible. However, real-world testing remains challenging due to the lack of infrastructure that would allow safe deactivation of the reachability constraint i.e., enabling potential collisions in a controlled manner. Other viable directions include reducing computational demands via learning-based off-line BRT approximations, performing sensitivity analysis on threshold parameters and extending the framework to scenarios involving cooperative interactions among multiple autonomous vehicles.

CRediT authorship contribution statement

Francesco Paparazzo: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization; **Marco Doria Fragomeni:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Mohammed Irshadh Ismaeel Sathyamangalam Imran:** Writing – review & editing, Visualization, Software, Data curation; **Stefano Arrigoni:** Writing – review & editing, Supervision, Project administration; **Francesco Braghin:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Ames, A. D., Coogan, S., Egerstedt, M., Notomista, G., Sreenath, K., & Tabuada, P. (2019). Control barrier functions: Theory and applications. <https://arxiv.org/abs/1903.11199>.
- Ammour, M., Orjuela, R., & Basset, M. (2022). A MPC combined decision making and trajectory planning for autonomous vehicle collision avoidance. *IEEE Transactions on Intelligent Transportation Systems*, 23(12), 24805–24817. <https://doi.org/10.1109/ITIS.2022.3210276>
- Bansal, S., Chen, M., Herbert, S., & Tomlin, C. J. (2017). Hamilton-jacobi reachability: A brief overview and recent advances. <https://doi.org/10.48550/arXiv.1709.07523>
- Bansal, S., & Tomlin, C. J. (2020). Deepreach: A deep learning approach to high-dimensional reachability. <https://doi.org/10.48550/arXiv.2011.02082>
- Başar, T., & Olsder, G. J. (1999). Dynamic Noncooperative Game Theory (vol. 23). Classics in Applied Mathematics. (2nd ed.). Philadelphia: Society for Industrial and Applied Mathematics. <https://doi.org/10.1137/1.9780898718620>
- Censi, A., Bolognani, S., Zanardi, A., Frazzoli, E., Liniger, A., & Schwager, M. (2021). Game theoretical motion planning. In *Icra 2021 - tutorial session*. <https://ethz.ch/content/dam/ethz/special-interest/mavt/dynamic-systems-n-control/idsc-dam/ResearchFrazzoli/workshops/ACGTMTutorial-Intro.pdf>.
- Chen, M. (2017). High Dimensional Reachability Analysis: Addressing the Curse of Dimensionality in Formal Verification. Ph.D. thesis. EECS Department, University of California, Berkeley. <https://doi.org/10.1109/EECS.2017.132>
- Chen, M., Herbert, S., & Tomlin, C. J. (2016). Exact and efficient hamilton-jacobi-based guaranteed safety analysis via system decomposition. <https://doi.org/10.48550/arXiv.1609.05248>
- Chen, M., & Tomlin, C. J. (2018a). Hamilton-jacobi reachability: Some recent theoretical advances and applications in unmanned airspace management. *Annual Review of Control, Robotics, and Autonomous Systems*, 1(Volume 1, 2018), 333–358. <https://doi.org/https://doi.org/10.1146/annurev-control-060117-104941>
- Chen, M., & Tomlin, C. J. (2018b). Hamilton-jacobi reachability: Some recent theoretical advances and applications in unmanned airspace management. *Annual Review of Control, Robotics, and Autonomous Systems*, 1(Volume 1, 2018), 333–358. <https://www.annualreviews.org/content/journals/10.1146/annurev-control-060117-104941>
- Chilakamuri, V. K., Feng, Z., & Bansal, S. (2024). Reachability analysis for black-box dynamical systems. <https://doi.org/10.48550/arXiv.2410.07796>
- Coskun, S. (2021). Autonomous overtaking in highways: A receding horizon trajectory generator with embedded safety feature. *Engineering Science and Technology, an International Journal*, 24, 1049–1058. https://www.researchgate.net/publication/350703793_Autonomous_overtaking_in_highways_A_receding_horizon_trajectory_generator_with_embedded_safety_feature.
- Cruz-Maya, A., & Europe, S. R. (2020). Target reaching behavior for unfreezing the robot in a semi-static and crowded environment. <https://arxiv.org/abs/2012.01206>.
- Dixit, S., Fallah, S., Montanaro, U., Dianati, M., Stevens, A., McCullough, F., & Mouzakitis, A. (2018). Trajectory planning and tracking for autonomous overtaking: State-of-the-art and future prospects. *Annual Reviews in Control*, 45, 76–86. <https://www.sciencedirect.com/science/article/pii/S136757881730130X>. <https://doi.org/https://doi.org/10.1016/j.arcontrol.2018.02.001>
- Dixit, S., Montanaro, U., Dianati, M., Oxtoby, D., Mizutani, T., Mouzakitis, A., & Fallah, S. (2020). Trajectory planning for autonomous high-speed overtaking in structured environments using robust MPC. *IEEE Transactions on Intelligent Transportation Systems*, 21(6), 2310–2323. <https://doi.org/10.1109/ITIS.2019.2916354>
- Evans, L. C., & Souganidis, P. E. (1984). Differential games and representation formulas for solutions of hamilton-jacobi-isaacs equations. *Indiana University Mathematics Journal*, 33(5), 773–797. <https://doi.org/10.1512/iumj.1984.33.33039>
- Fisac, J. F., Bronstein, E., Stefansson, E., Sadigh, D., Sastry, S. S., & Dragan, A. D. (2018). Hierarchical game-theoretic planning for autonomous vehicles. <https://arxiv.org/abs/1810.05766>.
- Fridovich-Keil, D., Ratner, E., Peters, L., Dragan, A. D., & Tomlin, C. J. (2019). Efficient iterative linear-quadratic approximations for nonlinear multi-player general-sum differential games. <https://arxiv.org/abs/1909.04694>.
- Garg, K., Usevitch, J., Breeden, J., Black, M., Agrawal, D., Parwana, H., & Panagou, D. (2023). Advances in the theory of control barrier functions: Addressing practical challenges in safe control synthesis for autonomous and robotic systems. <https://arxiv.org/abs/2312.16719>.
- Gurses, Y., Buyukdemirci, K., & Yildiz, Y. (2023). Developing driving strategies efficiently: A skill-based hierarchical reinforcement learning approach. <https://arxiv.org/abs/2302.02179>.
- Hegedüs, T., Nemeth, B., & Gaspar, P. (2020). Multi-objective trajectory design for overtaking maneuvers of automated vehicles. In *21st IFAC world congress* (pp. 2050–2055). <https://ifattwww.et.uni-magdeburg.de/ifac2020/media/pdfs/2050.pdf>.
- Herbert, S., Chen, M., Bansal, S., et al. (2020). helperOC: An optimal control toolbox for hamilton-jacobi reachability analysis. Accessed: 2025-03-11, <https://github.com/HJReachability/helperOC>.
- Khatib, O. (1986). Real-time obstacle avoidance for manipulators and mobile robots. *The International Journal of Robotics Research*, 5(1), 90–98. <https://doi.org/10.1177/027836498600500106>
- Khayyat, M., Zanardi, A., Arrigoni, S., & Braghin, F. (2024). Tactical game-theoretic decision-making with homotopy class constraints. <https://doi.org/10.1109/TIV.2024.3467701>
- Lattarulo, R., He, D., & Pérez, J. (2018). A linear model predictive planning approach for overtaking manoeuvres under possible collision circumstances. In *2018 IEEE intelligent vehicles symposium (IV)* (pp. 1340–1345). <https://doi.org/10.1109/IVS.2018.8500542>
- Leung, K., Schmerling, E., Zhang, M., Chen, M., Talbot, J., Gerdes, J. C., & Pavone, M. (2020). On infusing reachability-based safety assurance within planning frameworks for human-robot vehicle interactions. *International Journal of Robotics Research*, 39, 1326–1345. <https://doi.org/10.1177/0278364920946814>
- Li, N., Oyler, D. W., Zhang, M., Yildiz, Y., Kolmanovsky, I., & Girard, A. R. (2018). Game theoretic modeling of driver and vehicle interactions for verification and validation of autonomous vehicle control systems. *IEEE Transactions on Control Systems Technology*, 26(5), 1782–1797. <https://doi.org/10.1109/TCST.2017.2723574>
- Liu, J., Li, H., Yang, Z., Dang, S., & Huang, Z. (2022). Deep dense network-based curriculum reinforcement learning for high-speed overtaking. *IEEE Intelligent Transportation Systems Magazine*, . <https://doi.org/10.1109/ITS.2022.3174410>
- Lodhi, S. S., Kumar, N., & Pandey, P. K. (2023). Autonomous vehicular overtaking maneuver: A survey and taxonomy. *Vehicular Communications*, 42, 100623. <https://www.sciencedirect.com/science/article/pii/S2214209623000530>. <https://doi.org/https://doi.org/10.1016/j.vehcom.2023.100623>
- Lu, H., Lu, C., Yu, Y., Xiong, G., & Gong, J. (2022). Autonomous overtaking for intelligent vehicles considering social preference based on hierarchical reinforcement learning. *Automotive Innovation*, 5, 195–208. <https://doi.org/10.1007/s42154-022-00177-1>
- Mitchell, I. (2020). A robust controlled backward reach tube with (almost) analytic solution for two dubins cars. In G. Frehse, & M. Althoff (Eds.), *Arch20. 7th international workshop on applied verification of continuous and hybrid systems (arch20)* (pp. 242–258). (vol. 74). EPIc Series in Computing. <https://doi.org/10.29007/mx3f>
- Mitchell, I. M. (2004). A Toolbox of Level Set Methods. Technical Report UBC CS TR-2004-09. Department of Computer Science, University of British Columbia. Accessed: 2025-03-11, <https://www.cs.ubc.ca/~mitchell/ToolboxLS/toolboxLS-1.0.pdf>.
- Mitchell, I. M., Bayen, A. M., & Tomlin, C. J. (2005). A time-dependent hamilton-jacobi formulation of reachable sets for continuous dynamic games. *IEEE Transactions on Automatic Control*, 50(7), 947–957. <https://doi.org/10.1109/TAC.2005.851439>
- Németh, B., Hegedüs, T., & Gáspár, P. (2019). Model predictive control design for overtaking maneuvers for multi-vehicle scenarios. In *2019 18th european control conference (ECC)* (pp. 744–749). <https://doi.org/10.23919/ECC.2019.8796013>

- Ortega, J., Ortega, M., Ismael, K., Ortega, J., & Moslem, S. (2024). Systematic review of overtaking maneuvers with autonomous vehicles. *Transportation Engineering*, 17, 100264. <https://www.sciencedirect.com/science/article/pii/S2666691X24000393>. <https://doi.org/https://doi.org/10.1016/j.treng.2024.100264>
- Osher, S., & Fedkiw, R. (2003). Level Set Methods and Dynamic Implicit Surfaces (vol. 153). Applied Mathematical Sciences. Springer. <https://doi.org/10.1007/b98879>
- Paparazzo, F., Castoldi, A., Sathyamangalam Imran, M. I. I., Arrigoni, S., & Braghin, F. (2025). Learning-based MPC leveraging SINDy for vehicle dynamics estimation. *Electronics*, 14(10). <https://doi.org/10.3390/electronics14101935>
- Qian, H., Zhao, L., Hawbani, A., Liu, Z., Yu, K., He, Q., & Bi, Y. (2024). Collaborative overtaking strategy for enhancing overall effectiveness of mixed connected and connectionless vehicles. *IEEE Transactions on Mobile Computing*, 23(12), 13556–13572. <https://doi.org/10.1109/TMC.2024.3427677>
- Rezaee, K., Yadmellat, P., & Chamorro, S. (2021). Motion planning for autonomous vehicles in the presence of uncertainty using reinforcement learning. <https://arxiv.org/abs/2110.00640>
- Song, X., Gao, H., Ding, T., Gu, Y., Liu, J., & Tian, K. (2023). A review of the motion planning and control methods for automated vehicles. *Sensors*, 23(13), 6140. <https://doi.org/10.3390/s23136140>
- StanfordASL (2024). hj_reachability: Hamilton-jacobi reachability analysis in JAX. Accessed: 2025-03-11, https://github.com/StanfordASL/hj_reachability.
- Tariq, F. M., Suriarachchi, N., Mavridis, C., & Baras, J. S. (2022). Autonomous vehicle overtaking in a bidirectional mixed-traffic setting. In *2022 american control conference (ACC)* (pp. 3132–3139). <https://doi.org/10.23919/ACC53348.2022.9867432>
- Teng, S., Hu, X., Deng, P., Li, B., Li, Y., Yang, D., Ai, Y., Li, L., Xuanyuan, Z., Zhu, F., & Chen, L. (2023). Motion planning for autonomous driving: The state of the art and future perspectives. *IEEE Transactions on Intelligent Vehicles*, 8(1), 123–144. <https://doi.org/10.1109/TIV.2023.3274536>
- Verschueren, R., Frison, G., Kouzoupis, D., Frey, J., van Duijkeren, N., Zanelli, A., Novoselnik, B., Albin, T., Quirynen, R., & Diehl, M. (2021). acados – a modular open-source framework for fast embedded optimal control. *Mathematical Programming Computation*, . <https://doi.org/10.1007/s12532-021-00200-3>
- Wang, X., Leung, K., & Pavone, M. (2020). Infusing reachability-based safety into planning and control for multi-agent interactions. *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, (pp. 6252–6259). <https://api.semanticscholar.org/CorpusID:215415534>
- Wang, X.-F., Chen, W.-H., Jiang, J., & Yan, Y. (2024). High-level decision-making for autonomous overtaking: An MPC-based switching control approach. *IET Intelligent Transport Systems*, 18(7), 1259–1271. <https://doi.org/https://doi.org/10.1049/itr2.12507>
- Xie, J., Hu, L., Qin, J., Yang, J., & Gao, H. (2024). Safety-critical control with control barrier functions: A hierarchical optimization framework. <https://arxiv.org/abs/2410.15877>
- Xie, S., Hu, J., Arvin, F., Ding, Z., & Bhowmick, P. (2022). Distributed motion planning for safe autonomous vehicle overtaking via artificial potential field. *IEEE Transactions on Intelligent Transportation Systems*, 23(11), 21531–21547. <https://doi.org/10.1109/TITS.2022.3189741>
- Yang, J., Lee, S., Lim, W., & Sunwoo, M. (2021). Human-like decision-making system for overtaking stationary vehicles based on traffic scene interpretation. *Sensors*, 21(20). <https://doi.org/10.3390/s21206768>
- Yang, L., Lu, C., Xiong, G., Xing, Y., & Gong, J. (2022). A hybrid motion planning framework for autonomous driving in mixed traffic flow. *Green Energy and Intelligent Transportation*, 1(3), 100022. <https://doi.org/10.1016/j.geits.2022.100022>
- Yu, S., Chen, B., Jaimoukha, I. M., & Evangelou, S. A. (2024). Game-theoretic model predictive control for safety-assured autonomous vehicle overtaking in mixed-autonomy environment. In *2024 european control conference (ECC)* (pp. 3728–3733). <https://doi.org/10.23919/ECC64448.2024.10591086>
- Yuan, D., Yu, X., Li, S., & Yin, X. (2024). Safe-by-construction autonomous vehicle overtaking using control barrier functions and model predictive control. *International Journal of Systems Science*, 55(7), 1283–1303. <https://doi.org/10.1080/00207721.2024.2304665>
- Zanardi, A., Sessa, P. G., Käslin, N., Bolognani, S., Censi, A., & Frazzoli, E. (2023). How bad is selfish driving? bounding the inefficiency of equilibria in urban driving games. *IEEE Robotics and Automation Letters*, 8. <https://ieeexplore.ieee.org/document/10058179>
- Zhao, C., Li, L., Pei, X., Li, Z., Wang, F.-Y., & Wu, X. (2020). A comparative study of state-of-the-art driving strategies for autonomous vehicles. *Accident Analysis & Prevention*, 150, 105937. <https://doi.org/10.1016/j.aap.2020.105937>
- Zhou, X., Kuang, D., Zhao, W., Xu, C., Feng, J., & Wang, C. (2020). Lane-changing decision method based nash q-learning with considering the interaction of surrounding vehicles. *IET Intelligent Transport Systems*, 14(14), 2064–2072. <https://doi.org/https://doi.org/10.1049/iet-its.2020.0427>